

Designing Food Delivery Gig-platforms for Courier Needs: the Case of Batched Orders

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ABSTRACT

The food delivery gig-platforms such as Uber Eats and Deliveroo increasingly batch orders, i.e., couriers deliver multiple orders in one trip to optimize order delivery. Prior studies investigate the topic of batching orders from a theoretical perspective, i.e., optimization, without taking contextual elements and worker preferences into account. In our paper, we elaborate on the couriers' perspectives and needs in batched order delivery based on a qualitative study. We identify several criteria used by couriers to accept and reject the batched orders. We propose an alternative order batching solution that supports courier needs.

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1 INTRODUCTION

The food delivery apps like Uber Eats allow any restaurant to have their food delivered to customers' doors relying on an outsourced fleet of couriers through a platform that connects a customer and a restaurant to a courier. These apps, which we call gig-platforms, are organized following the gig-economy model. The field of gig-economy and gig-workers has been an important domain of research in HCI and CSCW communities [3, 6, 11, 15, 19]. Gig-workers appreciate autonomy and flexibility in their activity, but this is counter-balanced by several issues. These are: platformic or algorithmic management [7, 8, 18, 21]; incentivization to work for some time and place [21, 22]; the lack of transparency on the consequences of rating [5]; and unpredictable task assignment [5, 12]. The algorithmic management implies that the "app" defines all aspects of the work: task assignment and waiting times before new tasks, reviews, work assessment and pay. Studies revealed that many app

decisions were unclear to the workers, and they tended to ground the app decisions on rationality [4].

In the food delivery gig-platform, the couriers are self-employed workers involved as partners with the platform. The couriers are paid for every delivery, and they have a portion of their time at work that is not paid, for instance, when they wait for a new order. Their activity is completely tied to the platform's functioning, and their account can be deactivated without further explanation [13, 23]. The food delivery sector presents several challenges as investigated in prior literature. First, there is the dynamism and urgency of the arriving flow of orders. Some researchers present meal delivery as the "ultimate challenge in last mile logistics" as a standard order is expected to be delivered within an hour that also comprises the time to prepare the food [20]. Time is a key element in guaranteeing the quality of the meal delivered. Second, the orders in food delivery spike at peak hours. At these hours, demand is high, and the supply of couriers is limited. It leads to two difficult outcomes: either the delivery companies stop taking orders, or they have difficulties meeting the delivery promises [16]. Researchers have identified a need for efficient and effective batching technology to support meal delivery operations [1]. This includes batching of the orders as an answer to the constraints of the domain [9, 10, 16]. However, the references we point to on batching orders propose solutions from a very theoretical perspective, i.e., optimization. Contextual elements and worker preferences are largely unaccounted for in the design of such algorithms. Previous studies point to the necessity for such platforms to evolve toward fairer working conditions, better consideration of workers' voices, and more humane management [17, 21].

Platform-Supported Cooperative Work (PSCW) [2], a special interest group for CSCW and ECSCW researchers, investigates how platforms create new working conditions and cooperation. We aim to contribute to this growing field of research. In our paper, we investigate the understanding and needs of the couriers in the food delivery platforms and propose implications for platform design. We conducted semi-structured interviews and collected posts from courier forums to gather insights into the working conditions and platform-worker relationships. We found intense discussions regarding the multiple order assignment and delivery. Hence, in this paper, we focus on batched order.

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2 METHOD

We conducted 8 semi-structured interviews of couriers (6 male, 2 female, working with 1-3 gig-platforms), which were anonymized and transcribed. The goal was to have a broad understanding of the courier's activity on gig-platforms. We triangulated our findings by analyzing posts in two Facebook forums of couriers and 10 self-made videos of couriers openly shared online, as described in [12, 14]. We manually collected over 500 posts and comments over seven months from two private Facebook forums: "Forum of couriers 1" (11.1K members, approximately 100 posts/months) and "Forum of couriers 2" (31.7k members, approximately 270 posts/months). In this content, we extracted 51 posts and 905 comments manually on the topic of batched orders from the Facebook forums, and 2 out of 10 videos were on batching. Through the interviews, the posts, and the videos, we could build an understanding of the couriers' experiences, motivations, and needs. We analyzed all the content following an ethnomethodological perspective. In this contribution, we articulate the specific needs identified regarding batching orders.

3 FINDINGS

Our findings refer to how food delivery apps were used at the time of the study in France. We acknowledge that gig-platforms regularly update their app and differ geographically. During the interviews, all our participants mentioned that they were proposed batched orders by the platform and had performed batched orders delivery.

For all our participants, the most common motivation to be a courier was the flexibility of work and financial gain. For 4/8 of couriers, it was their main activity; for 2/8, it was a backup job till they found something else, and for 2/8, it was a supplementary job to have an additional income. Participants described delivery work as "irregular" (P2) and "a physical job" (P3). Orders were always assessed with a perspective of reward and effort. "If I see that it is a neighborhood where there is a big climb or if I see that it is 6km from my house and by bike I am going to waste half an hour doing that, it is not worth it the shot. Must try to be the most profitable." (P1)

3.1 (Lack of) Profitability of batched orders

Recently, multiple orders have appeared as the current norm for the interviewees: "Lately, almost all of them are doing double orders. It is rare where they do singles for us" (P5). There are two options: "Sometimes, the two orders come directly at the same time. On the notification, they write you two orders and two addresses, also the price and the total mileage. Sometimes you receive only one order, and then they tell you [for an additional one you can accept or refuse]" (P8). In both cases, couriers could cancel one of the orders only after accepting them at the time of the study. Batched orders were inflexible, and couriers couldn't change the delivery sequence.

The participants explained that the batched orders were meaningful when looking at the location criteria. "Either you have the two restaurants next to each other, and you are offered to take the order like that, or it is the same restaurant and people who live just next door, for example" (P3). However, there are mixed feelings regarding batched orders among gig-workers. There are moments

when the couriers perceive batched orders as attractive, i.e., during the bonus period. "The bonuses are if you make 6 deliveries within a period of time, so for example in 1h30 or in 2h. [...], you will have plus 1€50 on each order. So, if you made 6 orders at 3€ in less than 2h, instead of 18€, you have earned 27€" (P1). In those circumstances, "double orders are good because it allows you to accumulate two orders instead of one, so you can get the bonus faster" (P7). However, the effort to deliver multiple orders does not always translate to equal benefits, as captured in this post: "Let us stop accepting double orders outside of bonus periods!!! Choose 1 and cancel the other. Otherwise, it is like doing a free ride for Uber Eats every time, or just for 1€ or 2" (Forum of Couriers 1). Many couriers explained to us the reason why the batched orders were perceived as under priced: "If you take an order, let us say 2km, they will pay you 4.50€ and the second one will be just 500m after that one, it will be a distance of 2.5km. Instead of giving it to someone else at 5€, they will add it to you and make you +1.5€. It's a win for the platforms." (P5)

3.2 Increased risks in batched orders

Another concern regarding multiple orders is how the customers perceive the couriers and the effect of waiting time. "There are some questions I have about double orders. When we start to deliver the first customer, how are we traced at the "second customer"? Does the second customer see the route we take for the first delivery? Is the waiting time for the second customer calculated according to the distance between the first and the second customer?" (Forum of Couriers 1). Here the perceived risk for the courier lies in the possibility of a bad rating by customers if they are not well aware of the multiple order situation. It is further inflamed if there is a long wait in one of the two order pick-ups. "The first you pick it up in a first restaurant and the second in a second restaurant and there, the first you will find it ready, you go to the second restaurant, and you will stay 15-20min waiting while the first customer is already starting to message you to find out where his order is" (P5). The couriers also mentioned that they tended to refuse the orders from specific restaurants for two main reasons, order packaging and the typical wait times that they knew from experience. "It can be a bowl [fragile packaging] or... most of the time; it's because restaurants are running late." (P6). Other reasons to refuse orders include if the size of the order is too big or the orders are too far apart. "When the order is bigger than my bag, I cancel it. If the distance is too much, I do not want to go with the cycle, so I cancel it." (P4)

From these quotes, we understand that the three main reasons why couriers are reluctant to take batched orders are 1) they earn less than if they were to do both orders separately, 2) there is a risk in making the customer wait which can result in a bad rating and 3) there are other criteria such as time lost in waiting and specific order characteristics that algorithms cannot account for. We see that the interests of gig-workers, the couriers, are not necessarily aligned with the platform's interest and the courier's practices of accepting/rejecting for contextual reasons like wait time and local knowledge of a restaurant are not well reflected in the current platform designs.

3.3 Batching from different platforms

A solution for couriers to gain profitability is to combine single orders from different platforms. "You earn more [money] to deliver a double over two different applications than a double order from the same application" (Forum of couriers 2). It seems to be a common practice, as observed in both the self-made online videos and the interviews. It is especially practiced when there is a long wait for the order of one platform. "I do it very often with Uber Eats and Deliveroo" (P2), even if it is risky. P5 explained receiving a notification from Deliveroo reproaching him for not delivering the order directly, as they should do. "I had two orders [in the bag], one Uber and one Deliveroo, I started with Uber, and then I went for [the Deliveroo one]; they were on the same route, but I did not go directly [to deliver the Deliveroo order].[...] on the email, they say 'a customer reported that you did not leave (come to the address) directly'".

In conclusion, couriers decline batched orders from the same platform as they perceive it as unprofitable and risky. However, they are inclined to batch orders themselves among several platforms as it becomes profitable even though risky. This illustrates the worker's tendency to gain back agency even in such a constrained work context.

4 DISCUSSION

4.1 Implications for design

Our findings point to the fact that batching of orders is not sufficiently incentivized through income increase for the gig-couriers, nor are the worker's needs supported by the technology in gig-platforms. The conditions supporting the current working of gig-platforms are likely to evolve in the future. First, the predictions of continuous growth of the food delivery market might face a limitation in the over-supply of workers at some point. They will not be able to afford an unlimited number of gig-couriers and will have to optimize the delivery, for instance, relying heavily on the batching orders. Second, the legal framework will evolve in some countries constraining gig-platforms to provide fairer working conditions to gig-workers. Hence, including workers' needs in the design of gig-platforms will be imperative.

Our research identified different criteria that couriers take into account when accepting or rejecting a batched order. The most common is the location of pick-ups and drop-offs and their distance. Even if the pick-ups are close together, the couriers tend to reject the order if the drop-offs are far from each other. The next criteria are the time constraints that include maximum time for the delivery so that the food is still hot (or cold) and fresh and the waiting times at the restaurants. The platforms automatically compute the maximum time based on the average time to ride and the allocated time for pick-up. However, the gig-platforms do not account for the actual waiting times at the restaurant. Another criterion is the order characteristics, such as the order's size, fragility, and temperature. Some couriers tend to reject the orders if the order will not fit in their bag or if they do not have a compartment to keep the hot and cold food separate. Over time, couriers often know the habits of local restaurants, such as the size of their orders, packaging, and general waiting times. An algorithm can partially or fully handle some basic and objective criteria, such as locations,

temporal constraints, order characteristics, and waiting time to pick up the orders. However, others are much more difficult to grasp, such as the habits of restaurants.

We also identified the need to give more agency to couriers in their decision-making. In the current practice of multiple deliveries, a courier is given a pre-batched order by the algorithm. The courier can cancel one of the batched orders only after accepting both of them. Moreover, the delivery sequence is prescribed and cannot be adjusted or reversed by the courier. This rearrangement might be desirable in cases when there is a delay at the first order pick up. In the interest of couriers, it makes sense that they can batch the orders the way they want to. One way can be showing multiple orders to the couriers that can be batched together to create their delivery route. Through platform design, the couriers can have more agency in their work to execute the multiple orders. Our proposed design visualizes the identified criteria to facilitate courier decision-making and gives them the ability to rearrange their order.

4.2 Proposed solution

We are currently developing an alternative order batching solution for gig-platforms that we will evaluate with the couriers in the future. Our solution suggests different alternatives to batch one order with others by sliding through the order cards and rearranging the orders by clicking the "shuffle" button, as seen in Fig 1. The system recomputes the length of the trip, pick-up time, and drop-off time of order deliveries according to the batching and displays their geospatial information on the screen. The UI enables direct visibility and access to all the previously identified criteria for appropriate batching and decision making. In the top half of the screen in 1c, the courier can see his location and the location of pick-ups and drop-offs. In the lower half, i.e., in the cards, the courier can see the order characteristics of the combined orders, i.e., the size of the orders, the temperature of the food, and the waiting time at the restaurant. The waiting time is calculated based on the reported delay by the restaurant and other couriers. The application enables the couriers to report delays and cancel one of the existing orders in case of a delay. Furthermore, it dynamically notifies when the proposed batched delivery is not desirable due to the time limit.

Our proposed solution aims to rebalance the agency towards the couriers by designing for the self-management of orders in the delivery activity.

5 CONCLUSION AND FUTURE WORK

Our study suggests that the couriers have mixed feelings towards multiple order delivery for two main reasons. First, it is not sufficiently profitable and second, the needs of the couriers' are not supported. In our research, we identified several criteria, such as the waiting time, order characteristics, and habits of the restaurants that couriers use for decision-making, in addition to the location criteria. Moreover, couriers lack the ability to rearrange the batched orders. We propose an alternative design for batching that considers courier input and provides more agency for decision-making for multiple order delivery. We plan to evaluate our proposed solution with couriers through field experiments in our future work and include customers' perspectives regarding the tracking of batched orders.

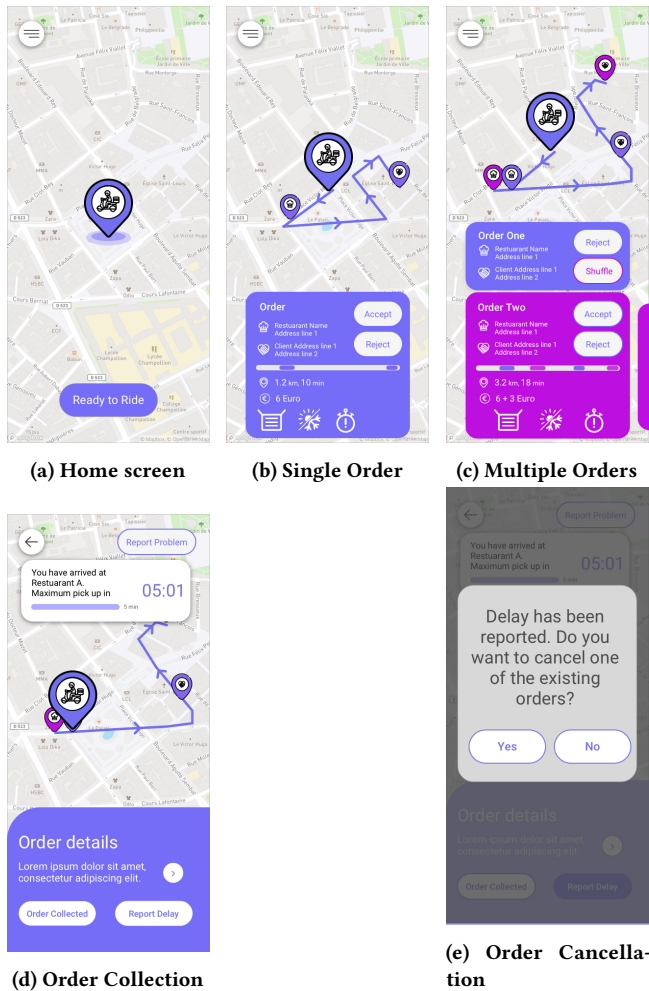


Figure 1: The interface of the proposed app. (a) is the home screen displaying the location of the courier. (b) is the single order interface. The card displays information such as the pick-up and drop-off address, the visual bar for estimated time, including waiting times, the total distance and time, and the total compensation. There are three symbols at the bottom; first, a bag icon representing the order size; second, the temperature of the order, whether it is hot, cold, or mixed; third, a clock that denotes normal or increased waiting time using tick or exclamation symbol within the clock. (c) is the double order interface with combined information for the batched orders in the cards. The couriers can batch orders the way they want from a set of four orders that they can swipe left to see and rearrange the delivery sequence. (d) shows the interface to report a delay and cancel the order if the waiting time at the pick-up is more than usual. (e) shows the option to cancel one of the existing orders.

REFERENCES

- [1] Xuetao Ding, Runfeng Zhang, Zhen Mao, Ke Xing, Fangxiao Du, Xingyu Liu, Guoxing Wei, Feifan Yin, Renqing He, and Zhizhao Sun. 2020. Delivery Scope: A New Way of Restaurant Retrieval for On-demand Food Delivery Service. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '20)*. Association for Computing Machinery, New York, NY, USA, 3026–3034. <https://doi.org/10.1145/3394486.3403353>
- [2] Babak A Farshchian, Miria Grisot, Stefan Hochwarter, Anna Sigríður Islind, Marius Mikalsen, Elena Parmiggiani, and Polyxeni Vassilakopoulou. 2021. Platform-Supported Cooperative Work. <https://www.eusset.eu/2021/11/02/platform-supported-cooperative-work/>
- [3] Mareike Glöss, Moira McGregor, and Barry Brown. 2016. Designing for Labour: Uber and the On-Demand Mobile Workforce. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*. Association for Computing Machinery, New York, NY, USA, 1632–1643. <https://doi.org/10.1145/2858036.2858476>
- [4] Karen Gregory. 2021. 'My Life Is More Valuable Than This': Understanding Risk among On-Demand Food Couriers in Edinburgh. *Work, Employment and Society* 35, 2 (April 2021), 316–331. <https://doi.org/10.1177/0950017020969593>
- [5] Benjamin Hanrahan, Ma Ning, and Yuan Chien Wen. 2017. The Roots of Bias on Uber. *Proceedings of 15th European Conference on Computer-Supported Cooperative Work - Exploratory Papers* (2017). <https://doi.org/10.1145/310420/ecs2017-27>
- [6] Lilly C. Irani and M. Six Silberman. 2013. Turkopticon: interrupting worker invisibility in amazon mechanical turk. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '13)*. Association for Computing Machinery, New York, NY, USA, 611–620. <https://doi.org/10.1145/2470654.2470742>
- [7] Mirela Ivanova, Joanna Bronowicka, Eva Kocher, and Anne Degner. 2018. *Foodora and Deliveroo: The App as a Boss? Control and autonomy in app-based management - the case of food delivery riders*. Working Paper 107. Working Paper Forschungs-förderung. <https://www.econstor.eu/handle/10419/216032>
- [8] Mohammad Hossein Jarrahi, Will Sutherland, Sarah Nelson, and Steve Sawyer. 2020. Platformic Management, Boundary Resources for Gig Work, and Worker Autonomy. *Computer Supported Cooperative Work* (April 2020). <https://doi.org/10.1007/s10606-019-09368-7>
- [9] Shenggong Ji, Yu Zheng, Zhaoyuan Wang, and Tianrui Li. 2019. Alleviating Users' Pain of Waiting: Effective Task Grouping for Online-to-Offline Food Delivery Services. In *The World Wide Web Conference (WWW '19)*. Association for Computing Machinery, New York, NY, USA, 773–783. <https://doi.org/10.1145/3308558.3313464>
- [10] Manas Joshi, Arshdeep Singh, Sayan Ranu, Amitabha Bagchi, Priyank Karia, and Puneet Kala. 2020. *Batching and Matching for Food Delivery in Dynamic Road Networks*. Technical Report arXiv:2008.12905 [cs] type: article. <http://arxiv.org/abs/2008.12905>
- [11] Aniket Kittur, Jeffrey V. Nickerson, Michael Bernstein, Elizabeth Gerber, Aaron Shaw, John Zimmerman, Matt Lease, and John Horton. 2013. The future of crowd work. In *Proceedings of the 2013 conference on Computer supported cooperative work (CSCW '13)*. Association for Computing Machinery, New York, NY, USA, 1301–1318. <https://doi.org/10.1145/2441776.2441923>
- [12] Min Kyung Lee, Daniel Kusbit, Evan Metsky, and Laura Dabbish. 2015. Working with Machines: The Impact of Algorithmic and Data-Driven Management on Human Workers. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI '15)*. Association for Computing Machinery, New York, NY, USA, 1603–1612. <https://doi.org/10.1145/2702123.2702548>
- [13] Ning F. Ma and Benjamin V. Hanrahan. 2019. Part-Time Ride-Sharing. *Proceedings of the ACM on Human-Computer Interaction* 3, GROUP (Dec. 2019), 1–17. <https://doi.org/10.1145/3361128>
- [14] David Martin, Benjamin V. Hanrahan, Jacki O'Neill, and Neha Gupta. 2014. Being a turker. In *Proceedings of the 17th ACM conference on Computer supported cooperative work & social computing (CSCW '14)*. Association for Computing Machinery, New York, NY, USA, 224–235. <https://doi.org/10.1145/2531602.2531663>
- [15] David Martin, Jacki O'Neill, Neha Gupta, and Benjamin V. Hanrahan. 2016. Turk-ing in a Global Labour Market. *Computer Supported Cooperative Work (CSCW)* 25, 1 (Feb. 2016), 39–77. <https://doi.org/10.1007/s10606-015-9241-6>
- [16] Ritwik Prashant Moghe, Sunil Rathee, Bharath Nayak, and Kranthi Mitra Adusumilli. 2020. Machine Learning based Batching Prediction System for Food Delivery. In *8th ACM IKDD CODS and 26th COMAD*. ACM, 316–322. <https://doi.org/10.1145/3430984.3430999>
- [17] Zane Muller. 2020. Algorithmic Harms to Workers in the Platform Economy: The Case of Uber. , 44 pages.
- [18] Mareike Möhlmann and Lior Zalmanson. 2017. *Hands on the wheel: Navigating algorithmic management and Uber drivers' autonomy*.
- [19] Noopur Raval and Paul Dourish. 2016. Standing Out from the Crowd: Emotional Labor, Body Labor, and Temporal Labor in Ridesharing. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing (CSCW '16)*. Association for Computing Machinery, New York, NY, USA, 97–107. <https://doi.org/10.1145/2818048.2820026>

- [20] Damian Reyes, Alan Erera, Martin Savelsbergh, Sagar Sahasrabudhe, and Ryan O'Neil. 2018. The Meal Delivery Routing Problem. (2018), 70.
- [21] Alex Rosenblat and Luke Stark. 2016. Algorithmic Labor and Information Asymmetries: A Case Study of Uber's Drivers. *International Journal of Communication* 10, 0 (July 2016), 27. <https://ijoc.org/index.php/ijoc/article/view/4892> Number: 0.
- [22] Gretchen M. Spreitzer, Lindsey Cameron, and Lyndon Garrett. 2017. Alternative Work Arrangements: Two Images of the New World of Work. *Annual Review of Organizational Psychology and Organizational Behavior* 4, 1 (2017), 473–499. <https://doi.org/10.1146/annurev-orgpsych-032516-113332> _eprint: <https://doi.org/10.1146/annurev-orgpsych-032516-113332>.
- [23] Alex Veen, Tom Barratt, and Caleb Goods. 2020. Platform-Capital's 'App-etite' for Control: A Labour Process Analysis of Food-Delivery Work in Australia. *Work, Employment and Society* 34, 3 (June 2020), 388–406. <https://doi.org/10.1177/0950017019836911> Publisher: SAGE Publications Ltd.